Arithmetic & Number Sense

ANSWER KEY

Section 1: Operations, Order of Operations & Decimals

1 What is the value of $18 - 3 \times (4 + 2) \div 2$?

a. 9 ✓

b. 12

c. 15

d. 6

Order of operations: Parentheses first: $4 + 2 = 6$. Multiplication and division from left to right: $3 \times 6 = 18$, then $18 \div 2 = 9$. Finally subtraction: $18 - 9 = 9$.

2 What is the value of $\frac{1}{0.2} + \frac{1}{0.5} + \frac{1}{0.25}$?

a. 11 ✓

b. 10

c. 9

d. 8

Convert decimals to fractions: $\frac{1}{0.2} = 5$, $\frac{1}{0.5} = 2$, $\frac{1}{0.25} = 4$. Summing gives $5 + 2 + 4 = 11$.

3 Evaluate the sum: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$.

a. 1 ✓

b. $\frac{5}{6}$

c. $\frac{7}{6}$

d. $\frac{4}{3}$

Find the common denominator 6: $\frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{6}{6} = 1$.

Section 2: Exponents, Radicals & Scientific Form

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 If $2^{x+3} = 64$, what is the value of x?

a. 3 ✓

b. 4

c. 5

d. 6

Since $64 = 2^6$, equating exponents yields $x + 3 = 6 \Rightarrow x = 3$.

5 Simplify the expression: $\sqrt{48} + \sqrt{27} - \sqrt{12}$.

a. $5\sqrt{3}$ ✓

b. $4\sqrt{3}$

c. $3\sqrt{3}$

d. $6\sqrt{3}$

Simplify each radical: $\sqrt{48} = 4\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$. Combining yields $4\sqrt{3} + 3\sqrt{3} - 2\sqrt{3} = 5\sqrt{3}$.

6 What is the value of $\frac{2^{10} - 2^9}{2^8}$?

a. 2 ✓

b. 4

c. 1

d. 8

Factor out 2^9 from the numerator: $\frac{2^9(2-1)}{2^8} = \frac{2^9(1)}{2^8} = 2^{9-8} = 2$.

Section 3: Divisibility Rules, Prime Factorization & Remainders

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 What is the units digit of 3^{42}?

a. 9 ✓

b. 3

c. 7

d. 1

Powers of 3 follow a 4-cycle of units digits: $3^1 \to 3, 3^2 \to 9, 3^3 \to 7, 3^4 \to 1$. Dividing $42 \div 4 = 10$ with remainder 2. Thus, the units digit is that of $3^2 = 9$.

8 The five-digit number $42a56$ is divisible by 9. What is the value of the digit a?

a. 1 ✓

b. 2

c. 0

d. 8

A number is divisible by 9 if the sum of its digits is a multiple of 9. Sum $= 4 + 2 + a + 5 + 6 = 17 + a$. The next multiple of 9 is 18, so $a = 18 - 17 = 1$.

9 What is the remainder when 4^{35} is divided by 5?

a. 4 ✓

b. 1

c. 2

d. 3

Notice the pattern of remainders modulo 5: $4^1 \equiv 4$, $4^2 = 16 \equiv 1$, $4^3 \equiv 4$. Since 35 is odd, the remainder is 4.

Section 4: Consecutive Integers & Arithmetic Reasoning

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 If the average of five consecutive integers is 24, what is the smallest integer?

a. 22 ✓

b. 20

c. 21

d. 23

For an odd count of consecutive integers, the average is the middle integer. Thus, the middle term is 24, and the five integers are $22, 23, 24, 25, 26$. The smallest is 22.

11 The sum of three consecutive odd integers is 75. What is the largest integer?

a. 27 ✓

b. 25

c. 29

d. 31

Let the integers be $n, n+2, n+4$. Then $3n+6=75 \Rightarrow 3n=69 \Rightarrow n=23$. The largest integer is $23+4=27$.

12 What is the greatest common divisor (GCD) of 48 and 72?

a. 24 ✓

b. 12

c. 18

d. 36

Prime factorizations: $48=2^4\times 3$ and $72=2^3\times 3^2$. The GCD takes the minimum powers: $2^3\times 3=8\times 3=24$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647